

CENG381 Stochastic Processes
Final Exam

Duration: 90 minutes

Q1 (30 points). Show a few steps of a 2D random walk starting at origin (0,0) with a discrete step size chosen from $[-1, 0, 1]$ with equal probability.

Q2 (35 points). Consider the Markov jump process on $S = \{1, 2, 3, 4\}$ with generator matrix

$$P = \begin{pmatrix} -1 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & -\frac{1}{2} & 0 & \frac{1}{4} \\ \frac{1}{6} & 0 & -\frac{1}{3} & \frac{1}{6} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

- a) Draw a transition rate diagram for this process.
- b) Write down the communicating classes for this process, and state whether they are recurrent or transient.
- c) Calculate the hitting probability h_{13} .
- d) Calculate the expected hitting time η_{14} .

Q3 (35 points). People arrive at a telephone booth in a random pattern with an average interarrival time of 12 minutes. The length of phone calls from the booth, including the dialing time, wrong numbers, etc. is exponentially distributed with an average time of 4 minutes.

- a) What is the probability that an arriving person will have to wait?
- b) What is the average length of the waiting lines that form from time to time; that is, those that are not of zero length?
- c) What is the probability that an arrival will have to wait for more than 10 minutes before the phone is available?